



Thermodynamics and transport properties of gabapentin, paracetamol, & albuterol in water

Ashok Chaudhary^{ID,1}, Bishnu Pandey^{ID,1}, Muskan Jalan^{ID}, Shyam Prakash Khanal^{*}, Narayan Prasad Adhikari^{ID,*}

Central Department of Physics, Tribhuvan University, Kirtipur, Kathmandu, 44600, Nepal

ARTICLE INFO

Keywords:

Molecular dynamics simulation
Free energy perturbation
Thermodynamic integration
Diffusion coefficient
Shear viscosity

ABSTRACT

The solubility and dissolution rates of drugs are significantly influenced by factors such as solvation-free energy and diffusion coefficient, which are essential for understanding their transport and thermodynamic properties. In this study, we investigated the free energy and transport properties of various drug molecules using classical molecular dynamics simulations. The systems were modeled using the OPLS force field. Solvation-free energies for paracetamol, gabapentin, and albuterol in water (TIP3P & TIP4P/2005 models) were computed at 310 K using thermodynamic integration (TI) and free energy perturbation (FEP) based methods across 21 coupling constant (λ) values. Additionally, solvent-accessible surface area (SASA) and hydrogen bond analyses provided further insights into the solvation behavior of these drugs. The diffusion coefficients were calculated using Einstein's and Darken's relations. The self-diffusion coefficients of the drugs and SPC/E water at 310 K showed good agreement with experimental data. Additionally, the shear viscosity of the drug-water solution has been studied using Einstein's relation. The structural properties of the system were analyzed using the radial distribution function (RDF) of different atomic pairs in solvent and solute. This work provides insights into the solvation and transport behavior of drug molecules, aiding in the understanding of their physicochemical properties.

1. Introduction

Gabapentin is a GABA-analog anticonvulsant [1] drug utilized in the treatment of peripheral neuropathic pains, postherpetic neuralgia, and partial-onset seizures [2]. It is 2-[1-(Aminomethyl)cyclohexyl]acetic acid with chemical formula: $C_9H_{17}NO_2$ [3]. The Gabapentin molecule consists of a γ -aminobutyric acid (GABA) substituted at the 1-position of cyclohexane [4] as shown in Fig. 1(a). Gabapentin reduces neuropathic pain by binding to the $\alpha_2\delta$ subunit [5] of voltage-gated calcium channels in the central nervous system [6], reducing calcium influx into neurons [7] and decreasing excitatory neurotransmitter release [8]. Gabapentin is absorbed in the intestine by LAT1 transporters and excreted through the renal method as an unchanged drug [9].

Acetaminophen, often known as paracetamol [10], is an antipyretic/analgesic medication used to treat fever and associated pain [11]. Paracetamol is 4-hydroxy acetanilide, N-acetyl-p-aminophenol having formula: $C_8H_9NO_2$ [12]. Paracetamol comprises a benzene ring with one hydroxyl group and one amide group in the (1,4) para position. The

two activation groups on the benzene ring make it extremely reactive to electrophilic substitution [13]. Fig. 1(b) shows the VMD snapshot of the molecular structure of the paracetamol. Paracetamol raises the pain threshold by blocking the production of prostaglandin (PG) [14] by two isoforms of Cyclooxygenase (COX), COX-1, and COX-2 [15]. The antipyretic effects of acetaminophen are most likely due to its direct impact on the brain's heat-regulating areas, which cause peripheral vasodilation, sweating, and heat loss [16]. The hydrogen bonds of paracetamol in its crystal form have been studied using the density functional theory approach [17]. The geometrical structure of paracetamol has been measured by neutron diffraction and X-ray diffraction techniques [18].

Albuterol, a short-acting β_2 -agonists (SABA) is used to treat bronchiectasis, chronic obstructive pulmonary disease, and asthma [19,20] Albuterol is considered to be among the safest and most effective medications that are vital to healthcare systems by the World Health Organization (WHO) [21–23]. The molecular formula of albuterol is $C_{13}H_{21}NO_3$ [24] and chemically known as 4-[2-(tert-Butylamino)-1-hydroxyethyl]-2-(hydroxymethyl)phenol. Albuterol is composed of a

* Corresponding author.

E-mail addresses: ashok.775511@cdp.tu.edu.np (A. Chaudhary), bishnu1.775511@cdp.tu.edu.np (B. Pandey), muskan.775511@cdp.tu.edu.np (M. Jalan), shyamkhanal1989@gmail.com (S.P. Khanal), narayan.adhikari@cdp.tu.edu.np (N.P. Adhikari).

¹ The authors contribute equally to this work.

<https://doi.org/10.1016/j.molliq.2025.127654>

Received 7 January 2025; Received in revised form 13 April 2025; Accepted 23 April 2025

Available online 28 April 2025

0167-7322/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

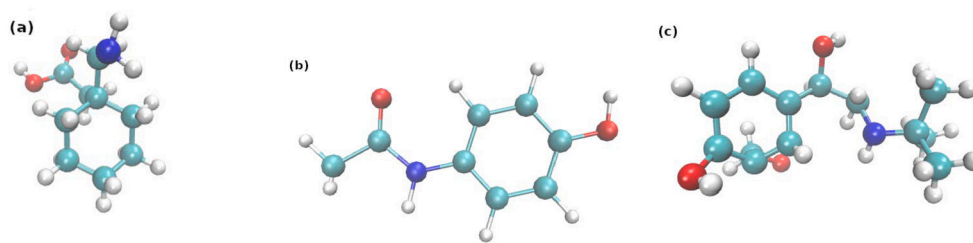


Fig. 1. VMD snapshot of (a) Gabapentin, (b) Paracetamol, and (c) Albuterol.

racemic mixture of the (R)- and (S)-enantiomers, with only the (R)-enantiomer responsible for its therapeutic effects [25,26]. The drug acts by binding to β_2 -receptors in the respiratory tract's smooth muscle, stimulating adenyl cyclase, and increasing cyclic AMP production, which leads to bronchial muscle relaxation and bronchodilation [27–30].

Several experimental and computational research studies have been conducted to understand the binding mechanism, interaction mechanics, solubility prediction, and optimization of the drug design [31–33]. The solvation-free energy is a fundamental thermodynamic quantity that plays a crucial role in various drug behaviors such as solubility, binding affinity, stability, and permeability [34–36]. The thermodynamic and transport properties of amoxicillin in water and ethanol have been studied. The free energy of solvation of amoxicillin was reported to be higher in water than ethanol by ≈ 105 kJ/mol [37]. In a study of the thermodynamic and transport properties of valine and cysteine peptides in water, the solvation-free energies of both peptides increased with chain length. In contrast, the diffusion coefficient of both valine and cysteine decreased with increasing monomer units [38]. The study of the adsorption of paracetamol in Boron Nitride Nanotube (BNNT) in different lengths and diameters of the nanotube [11]. Using the umbrella sampling method, MD simulations of paracetamol in DPPC and DMPC lipid bilayers showed a significant reduction in free energy at the bilayer center [39]. A study using MD simulations and the extended Einstein's crystal technique calculated the solubility of paracetamol in ethanol, finding the result ($X_{calc} = 0.085 \pm 0.014$) to be in close agreement with experimental value ($X_{exp} = 0.0585 \pm 0.004$) [40]. The study of phase behavior and thermodynamical stability of gabapentin anhydrate and monohydrate forms using different spectroscopic and computational approaches showed that the solubility trend of both forms of gabapentin was different with anhydrate exhibiting maximum solubility [41].

Similarly, the study of diffusion and viscosity [42] offers valuable insights into predicting a drug's mobility, interaction with other solutes and solvents, and absorption rates [43,44]. Diffusion is the key factor in the Noyes-Whitney equation [45] which helps to predict the dissolution rates. MD simulation is an effective tool to study these transport properties [46].

The precise calculation of the solvation-free energy of a molecule is crucial in various biomedical, chemical, and industrial research fields, and assessing it remains a longstanding and continually advancing obstacle in computational chemistry [34,47,48]. Some of the key applications are in drug discovery and development [49], material science [50], and environmental science [51]. Several computational methods are used to calculate the free energy. Some common methods are explicit solvent methods (molecular dynamics simulation [52] and Monte Carlo simulation [53]), implicit solvent methods (continuum solvation models [54]), and hybrid methods (quantum methods). Other methods include the free energy perturbation method, thermodynamic integration [37], and umbrella sampling [55]. The choice of method depends on the specific system and the desired level of accuracy.

In this work, we have presented the estimation of the solvation-free energy of gabapentin, paracetamol, and albuterol in water at a temperature of 310 K using classical MD simulation. In addition to this, we have also estimated the self & binary diffusion coefficient and viscosity of these drugs in water.

2. Theory

2.1. Solvation-free energy

The term “solvation-free energy” refers to the difference in the solute's free energy when it is in the gas phase (by itself) and when it is in a solution with solvent molecules around it [56,57]. Estimating the difference in free energy between two thermodynamic states plays a significant role in understanding various processes, including the functioning of drugs, protein-ligand interactions, and protein folding [58,59]. The total free energy of solvation was computed using thermodynamic integration and free energy perturbation.

2.1.1. Free energy perturbation

Free Energy Perturbation (FEP) is a computational technique used to evaluate the free energy differences between two molecular systems that differ by a small, incremental change in their structure or environment [60]. We explore the issue of changing a system from one thermodynamic state to another before starting our discussion of free energy differences. Consider these states are denoted by A and B, and characterized by potential energy functions $U_A(r_1, \dots, r_N)$ and $U_B(r_1, \dots, r_N)$ respectively [61]. The free energy is given by

$$\Delta F_{AB} = -k_B T \ln \langle \exp \{ -\beta (U_B - U_A) \} \rangle_A \quad (1)$$

Here, $\beta = \frac{1}{k_B T}$, and $\langle \dots \rangle_A$ is the ensemble average taken concerning the canonical configurational distribution of the state A.

For large values of $U_B - U_A$, the exponential factor $e^{-\beta(U_B - U_A)}$ becomes negligibly small which leads to slow convergence of energy. Due to this reason, this is useful only when two states A and B do not differ significantly. To overcome this problem of convergence, introduce set of M-2 intermediate states with potential $U_\alpha(r_1, r_2, \dots, r_N)$, where $\alpha = 1, \dots, M$. $\alpha = 1$ corresponds to A and $\alpha = M$ corresponds to state B. Thus free energy becomes:

$$\Delta F_{AB} = -k_B T \sum_{\alpha=1}^{M-1} \ln \langle e^{-\beta \Delta U_{\alpha, \alpha+1}} \rangle_\alpha \quad (2)$$

2.1.2. Thermodynamic Integration (TI)

The potential energy of merely two coordinate sets cannot be used to directly determine the free energy difference between two states. The free energy difference can be determined via TI by establishing a thermodynamics path between the states and integrating over ensemble-averaged changes along the path (TI). This method involves switching the system from one state to another.

We define a new potential energy function with switching parameter λ ,

$$U(r_1, \dots, r_N, \lambda) \approx f(\lambda)U_A(r_1, \dots, r_N) + g(\lambda)U_B(r_1, \dots, r_N) \quad (3)$$

The functions $f(\lambda)$ and $g(\lambda)$ are known as switching functions which should satisfy the following conditions $f(0) = 1$, $f(1) = 0$, $g(0) = 0$, and $g(1) = 1$. The free energy difference ΔF_{AB} can be obtained from the relation;

$$\Delta F_{AB} = \int_0^1 \left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda} d\lambda \quad (4)$$

In general, a set of M values of λ is chosen from the interval of $[0,1]$, and at each chosen λ_k a full molecular dynamics or Monte Carlo calculation is carried out to generate $\left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda_k}$. The obtained value of $\left\langle \frac{\partial U}{\partial \lambda} \right\rangle_{\lambda_k}$, $k=1, \dots, M$ are then substituted into the equation (3), and the result is integrated numerically to produce free energy difference ΔF_{AB} [61,62].

2.2. Diffusion coefficient

Diffusion is the process of movement of particles from an area of high concentration to an area of low concentration [63]. Self-diffusion coefficient is a quantitative measure of how fast a molecule of a substance moves randomly within its own structure due to thermal energy [64]. The Einstein relation is a well-known mathematical equation for calculating the self-diffusion coefficient from molecular positions which is given as [65,66],

$$D = \lim_{t \rightarrow \infty} \frac{\langle [r_{\alpha}(t + t_0) - r_{\alpha}(t_0)]^2 \rangle}{6t} \quad (5)$$

Here, α represents the type of components whether solute or solvent and t_0 is the initial time at origin, $\langle \dots \rangle$ represents the ensemble average [67].

The binary diffusion coefficient refers to the rate at which one component of a binary mixture diffuses through another component under the influence of a concentration gradient. The binary diffusion coefficient is computed by using Darken's relation [68], which is given as,

$$D_{AB} = N_B D_A + N_A D_B \quad (6)$$

Here, D_A and D_B represent the self-diffusion coefficients of species A and B respectively, and N_A and N_B are the corresponding mole fractions.

2.3. Shear viscosity

Shear viscosity, also known as apparent viscosity, is the shear stress applied to a fluid divided by the shear rate, or the relationship viscosity versus shear rate. This relationship is constant for Newtonian fluids and is known as Newton's Law of Viscosity [69]. The viscosity can be calculated from an equilibrium simulation using an Einstein relation [42]:

$$\eta = \frac{1}{2} \frac{V}{k_B T} \lim_{t \rightarrow \infty} \frac{d}{dt} \left\langle \left(\int_{t_0}^{t_0+t} P_{xz}(t') dt' \right)^2 \right\rangle_{t_0} \quad (7)$$

Here, in equation (7) V , T , k_B , and $P_{xz}(t')$ represent volume, temperature, Boltzmann constant, and pressure component (stress) respectively. $\langle \dots \rangle$ denotes the average over time origins. This method converges rather slowly and requires a very small number of steps. The result is very dependent on the treatment of the electrostatics. Using a (short) cut-off results in large noise on the off-diagonal pressure elements, which can increase the calculated viscosity by an order of magnitude [42,70].

3. Methodology

3.1. Modeling the system

In this work, we used GROMACS software for classical MD simulation [71]. The OPLS (Optimized Potentials for Liquid Simulations) force field [72] was used to model the system. The Lennard-Jones (LJ) and Coulomb potential were considered for the intermolecular interaction. The total intermolecular potential is given by:

$$U(r_{ij}) = 4\epsilon \left[\left(\frac{\sigma}{r_{ij}} \right)^{12} - \left(\frac{\sigma}{r_{ij}} \right)^6 \right] + \frac{q_i q_j}{4\pi\epsilon_m r_{ij}} \quad (8)$$

where r_{ij} , ϵ , σ , ϵ_m , q_i , and q_j are the distance between i^{th} and j^{th} atom, the strength of LJ potential, the distance at which LJ interaction is zero, the permittivity of the medium, the charge of i^{th} and j^{th} atom respectively [73].

3.2. Simulation details

3.2.1. Estimation of free energy

To estimate the free energy of solvation, we performed an MD simulation by mixing one gabapentin molecule with 677 water molecules (system I), one paracetamol with 999 water molecules (system II), and one albuterol with 1140 water molecules (system III) inside a cubic simulation box. The simulations were carried out at a temperature of 310 K and 1 atm pressure under a periodic boundary condition. The simulation used the OPLS force field [72] and two different water models: the TIP3P [74] and TIP4P/2005 [75] water models. In our simulations, we implemented a 1.2 nm cutoff for both non-bonded Coulomb and van der Waals interactions [38]. This distance is adequate for any cutoff method, provided that the cutoff radius does not exceed half the box size. Energy minimization was performed to minimize the atoms' unnatural overlapping and to adjust the solvent molecules' distribution and force field structure [73]. In our work, energy minimization was performed using the steepest descent algorithm (SD) with a tolerance value of $300 \text{ kJ mol}^{-1} \text{ nm}^{-1}$ and a position step size of 0.01 nm [66]. Equilibration was carried out to stabilize the system's temperature, pressure, and density before production runs. An equilibration run was performed using two thermodynamic ensembles (NVT and NPT) for 20 nanoseconds with a time step of 2 femtoseconds. Temperature coupling was achieved using a Langevin thermostat, while pressure coupling was achieved using a Parrinello-Rahman barostat with a coupling time of 1 picosecond. Long-range interactions are handled using the Particle Mesh-Ewald algorithm (PME) of order 6, with a Fourier spacing of 0.12. Hydrogen bonds were constrained using the LINCS algorithm [76]. After reaching the thermal and pressure equilibrium, a production run was carried out for 20 ns with a time step of 0.002 ps. A stochastic dynamics integrator was used for solvent molecule movement. Adjustments were made to softcore parameters, including $\text{sc-alpha}=0.5$, $\text{sc-power}=1$, and $\text{sc-sigma}=0.3$. All the other steps were the same as in the equilibration step. We have changed the non-bonded vdW and Coulomb interactions to define the intermediate states between the initial and final states. We defined 21 evenly spaced values for the coupling parameters of both Lennard-Jones (LJ) and Coulomb interact to transform the system from the initial state to the final state. The coupling parameter λ were: for Coulomb interaction λ_{Coulomb} : 0.00, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00, 1.00 and for van der Waals interaction λ_{vdW} : 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.00, 0.10, 0.20, 0.30, 0.40, 0.50, 0.60, 0.70, 0.80, 0.90, 1.00. After the production run for each value of λ , we used a Python scripted alchemical analysis [77] to estimate the free energy difference from the trajectories of the production run.

3.2.2. Transport properties

To estimate diffusion coefficients, we have utilized the SPC/E water model [78,79] with a similar system setup as that of solvation-free energy. First, energy minimization was performed using the steepest descent method to estimate diffusion. After the energy minimization, equilibration was carried out using an NPT ensemble for 0.5 μs simulation time with a time step of 2 fs. Throughout the simulations, electrostatic interactions were handled using Particle Mesh Ewald (PME) with a sixth-order interpolation, and a 1.0 nm cutoff was used for both short-range LJ and Coulomb interactions. Thermal equilibrium at temperature 310 K was achieved using a Velocity rescale thermostat and pressure equilibrium using a Brendsen barostat under the periodic boundary condition

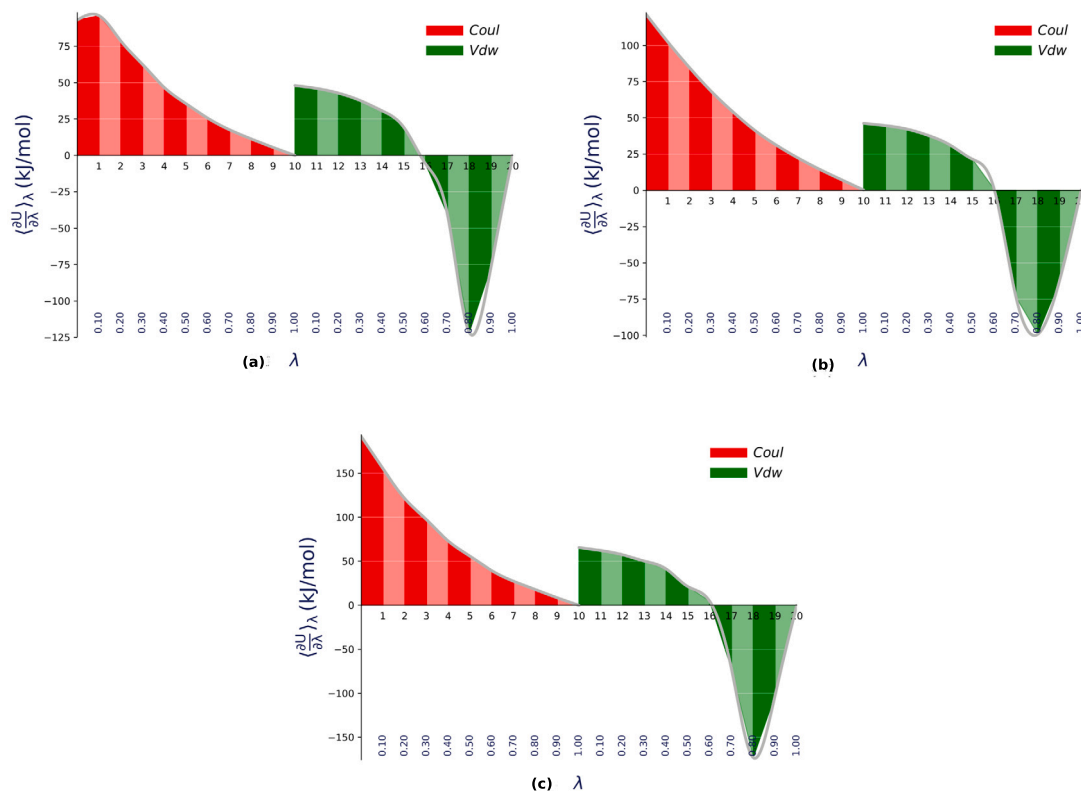


Fig. 2. Variation of $\left\langle \frac{\partial U}{\partial \lambda} \right\rangle_\lambda$ as a function of λ at 310 K temperature taking TIP3P water as solvent for (a) Gabapentin, (b) Paracetamol, and (c) Albuterol.

(PBC). All bonds were constrained by using the LINCS algorithm [80]. Finally, we performed the production run for a simulation time of 1 μ s with a time step of 2 fs.

4. Results and discussion

In this section, we present and discuss the findings of the present work. We have studied the free energy of solvation of Gabapentin, Paracetamol, and Albuterol in water along with hydrogen bonds and SASA. Further to elucidate the transport properties of the system, we have estimated self and binary diffusion coefficient and shear viscosity.

4.1. Free energy of solvation

Using an Alchemical analysis [77], the free energy of solvation of Gabapentin, Paracetamol, and Albuterol have been calculated from the production run trajectory for each lambda value. Solvation-free energies are estimated by transferring drug molecules from the vacuum state to the solvent (water) at 310 K temperature through MD simulation. We employed two techniques based on thermodynamic integration: TI and TI-CUBIC [61]. The two approaches are identical except that TI and TI-CUBIC use trapezoidal and cubic spline rules for numerical integration, respectively. Additionally, two methods based on the FEP method are employed: the Bennett Acceptance Ratio (BAR) and the Multistate Bennett Acceptance Ratio (MBAR) [62,81]. To estimate the free energy of solvation of all these systems, we introduced 21 distinct states, including the final state in which there is no interaction between the solute and solvent, and the initial state in which there is complete interaction between them through non-bonded vdW and Coulomb interactions [77]. The Variation of $\left\langle \frac{\partial U}{\partial \lambda} \right\rangle_\lambda$ as a function of λ has been plotted as shown in Fig. 2(a–c) for Paracetamol, Gabapentin, and Albuterol respectively.

The silver curve (color online) on the edge of the graphs in Fig. 2(a–c) represents interpolation using cubic splines (TI-CUBIC), while the filled area in each plot represents free energy estimations using the trapezoid rule (TI). The red and green colors indicate the contributions of Coulomb and vdW interactions to the solvation-free energy. The free energy difference between two successive coupling states is initially high but decreases as the coupling progresses. This occurred due to our initial consideration of both vdW and Coulomb interactions at full strength. We gradually turned off the interactions, first the vdW interaction and then the Coulomb interaction. In the Figs. 2(a–c), lambda states 1–10 represent the gradual turning off of Coulomb interactions while keeping van der Waals interactions constant. In lambda states 11–20, the van der Waals interactions slowly decrease while maintaining the Coulomb interactions constant. This illustrates the sequential decoupling of the two interaction components. The solvation-free energy of gabapentin, paracetamol, and albuterol is the sum of the free energy differences between the solute-solvent coupling in the initial state and the decoupling in the final state. Table 1 displays the estimated free energy of solvation values of Gabapentin, Paracetamol, and Albuterol in TIP3P water at 310 K using four distinct methods: TI, TI-CUBIC, BAR, and MBAR. The total free energy values are very close, with slight variations between methods. For Gabapentin, the total free energy ranges from -38.12 ± 0.13 (TI) to -38.39 ± 0.10 (MBAR), showing minimal differences. Similarly, for Paracetamol, free energy values range from -45.10 ± 0.09 (TI) to -44.42 ± 0.07 (BAR) which shows reasonable agreement with the experimental value ($\Delta G = -62.02 \pm 4.38$ [82]) considering differences in simulation conditions, water models, and the limitations inherent to force field-based molecular dynamics simulations. Albuterol's total free energy values are also consistent, ranging from -61.15 ± 0.14 (TI) to -60.18 ± 0.16 (MBAR). This supports the reliability and consistency of the solvation energy estimates from these above-mentioned computational approaches. From Table 1, it can be seen that solvation-free energy values are increasing with the size of the system. This implies that the energy required for solvation rises as the system size grows due

Table 1

Estimated values of free energy of solvation (ΔG_{solv}) in kJ/mol for Gabapentin, Paracetamol, and Albuterol in water (TIP3P and TIP4P/2005 models) at 310 K using TI, TI-CUBIC, BAR, and MBAR methods taking the individual contribution of vdW and Coulomb along with total contributions due to both the interactions.

Drugs	Water Model	Method	ΔG_{solv} in kJ/mol		
			vdW only	Coulomb only	Total
Gabapentin	TIP3P	TI	4.42 $\pm$ 0.09	-42.54 $\pm$ 0.09	-38.12 $\pm$ 0.13
		TI-CUBIC	5.06 $\pm$ 0.09	-42.65 $\pm$ 0.08	-37.59 $\pm$ 0.12
		BAR	5.04 $\pm$ 0.08	-43.47 $\pm$ 0.02	-38.43 $\pm$ 0.10
		MBAR	5.08 $\pm$ 0.05	-43.47 $\pm$ 0.03	-38.39 $\pm$ 0.10
	TIP4P/2005	TI	6.57 $\pm$ 0.20	-45.33 $\pm$ 0.05	-38.75 $\pm$ 0.20
		TI-CUBIC	7.26 $\pm$ 0.20	-45.24 $\pm$ 0.05	-37.98 $\pm$ 0.20
		BAR	7.08 $\pm$ 0.17	-45.22 $\pm$ 0.03	-38.14 $\pm$ 0.18
		MBAR	7.11 $\pm$ 0.20	-45.21 $\pm$ 0.05	-38.09 $\pm$ 0.21
Paracetamol	TIP3P	TI	3.44 $\pm$ 0.09	-48.53 $\pm$ 0.02	-45.10 $\pm$ 0.09
		TI-CUBIC	4.00 $\pm$ 0.09	-48.43 $\pm$ 0.02	-44.43 $\pm$ 0.09
		BAR	3.74 $\pm$ 0.06	-48.44 $\pm$ 0.02	-44.69 $\pm$ 0.07
		MBAR	3.83 $\pm$ 0.08	-48.46 $\pm$ 0.02	-44.62 $\pm$ 0.08
	TIP4P/2005	TI	2.02 $\pm$ 0.17	-49.46 $\pm$ 0.05	-47.44 $\pm$ 0.18
		TI-CUBIC	2.58 $\pm$ 0.17	-49.35 $\pm$ 0.05	-46.77 $\pm$ 0.18
		BAR	2.51 $\pm$ 0.14	-49.35 $\pm$ 0.03	-46.84 $\pm$ 0.14
		MBAR	2.58 $\pm$ 0.17	-49.36 $\pm$ 0.04	-46.78 $\pm$ 0.18
Albuterol	TIP3P	TI	7.83 $\pm$ 0.11	-69.98 $\pm$ 0.08	-61.15 $\pm$ 0.14
		TI-CUBIC	8.67 $\pm$ 0.11	-68.74 $\pm$ 0.12	-60.07 $\pm$ 0.16
		BAR	8.75 $\pm$ 0.10	-68.86 $\pm$ 0.09	-60.11 $\pm$ 0.13
		MBAR	8.79 $\pm$ 0.11	-68.97 $\pm$ 0.11	-60.18 $\pm$ 0.16
	TIP4P/2005	TI	6.81 $\pm$ 0.23	-74.81 $\pm$ 0.22	-68.00 $\pm$ 0.32
		TI-CUBIC	7.71 $\pm$ 0.22	-74.55 $\pm$ 0.25	-66.84 $\pm$ 0.33
		BAR	7.92 $\pm$ 0.23	-74.45 $\pm$ 0.13	-66.53 $\pm$ 0.26
		MBAR	7.96 $\pm$ 0.26	-74.48 $\pm$ 0.15	-66.52 $\pm$ 0.30

to enhanced solute-solvent interactions and possibly a higher degree of solvent structuring around larger solutes.

To investigate the effect of different water models on solvation-free energy calculations, we compared results obtained from the TIP3P and TIP4P/2005 models (Table 1). The solvation-free energy values were systematically higher with TIP4P/2005 than TIP3P for all three drug molecules, indicating stronger solute-solvent interactions in TIP4P/2005. This difference arises from the structural and electrostatic differences between the two water models [74,75]. While TIP3P places the negative charge on the oxygen atom, TIP4P/2005 positions it on a separate massless site, improving the representation of the water dipole moment and enhancing hydrogen bonding. Consequently, TIP4P/2005 leads to stronger electrostatic interactions and a more favorable solvation environment. This effect is further reflected in the individual contributions of van der Waals (vdW) and Coulombic interactions, where TIP4P/2005 exhibits higher Coulombic contributions and lower vdW contributions compared to TIP3P. For instance, in the case of Paracetamol, the Coulombic contribution changes from -48.53 ± 0.02 kJ/mol (TIP3P, TI) to -49.46 ± 0.05 kJ/mol (TIP4P/2005, TI), emphasizing the role of TIP4P/2005 in capturing stronger electrostatic interactions.

The free energy difference between two thermodynamic states has been determined through trajectory analysis. For reasonable estimation, the trajectory must be sampled at equilibrium to ensure convergence of the computed value [77,83]. To evaluate the convergence, the free energy difference has been plotted as a function of simulation time for both the forward and time-reversed data as shown in Fig. 3(a-c). During the forward process, the system moves from an initial state with $\lambda = 0$ to a final state with $\lambda = 1$. This entails gradually activating interactions between molecules. Conversely, in the reverse process, the system shifts from the final state with $\lambda = 1$ to the initial state with $\lambda = 0$. The interactions or parameters are gradually switched off or reversed in this case. Here in the graph, it is seen that the forward and reverse processes are following different paths for some fraction of a second. This is due to the hysteresis effect.

4.2. Hydrogen bonds and Solvent Accessible Surface Area (SASA)

H-bond analysis is an effective method for interpreting intricate molecular relationships and providing quantitative predictions regarding chemical and biological systems' dynamics, stability, structure, and function. In this work, we used trajectory after 20 ns production for the initial coupling state, denoted by $\lambda = 0$, to analyze the hydrogen bonds between solute and solvent molecules. We set cut-off parameters for length and angle during hydrogen bond analysis at 30° and 0.35 nm, respectively [37,73]. Fig. 4 shows the plots of the time evolution of the number of hydrogen bonds between solutes (Gabapentin, Paracetamol, and Albuterol) and solvent (TIP3P water) at 310 K temperature. Throughout the simulation, the graph indicates that the number of hydrogen bonds varies with an average value of roughly 4 for both Gabapentin and Paracetamol and 6 for Albuterol, highlighting stronger solute-solvent interactions in the Albuterol. This trend reflects differences in molecular structure and the accessibility of polar functional groups. In Gabapentin, the primary amine group ($-\text{NH}_2$) primarily acts as a hydrogen bond donor, while the carboxylic acid ($-\text{COOH}$) and nitrogen atoms can serve as acceptors. The orientation and spacing of these groups contribute to moderate hydration. Paracetamol, with its phenolic $-\text{OH}$ and amide ($-\text{NH}$ and $\text{C}=\text{O}$) groups, forms hydrogen bonds through both donor and acceptor sites, but steric hindrance and its planar structure reduce the overall accessibility of these functional groups. In contrast, Albuterol exhibits enhanced hydrogen bonding due to its multiple hydroxyl groups, which readily donate hydrogen bonds, and its ether and hydroxyl oxygen atoms, which function effectively as acceptors. The nature of solute-solvent interactions, including hydrogen bonding and surface accessibility, has also been explored in previous experimental studies [93], supporting our computational observations. As the number of hydrogen bonds increases, the solvation-free energy becomes more negative. This is because each hydrogen bond contributes to a stronger solute-solvent interaction, enhancing the solvation process. This Supports the higher value of estimated free energy of solvation for Albuterol followed by Paracetamol and Gabapentin.

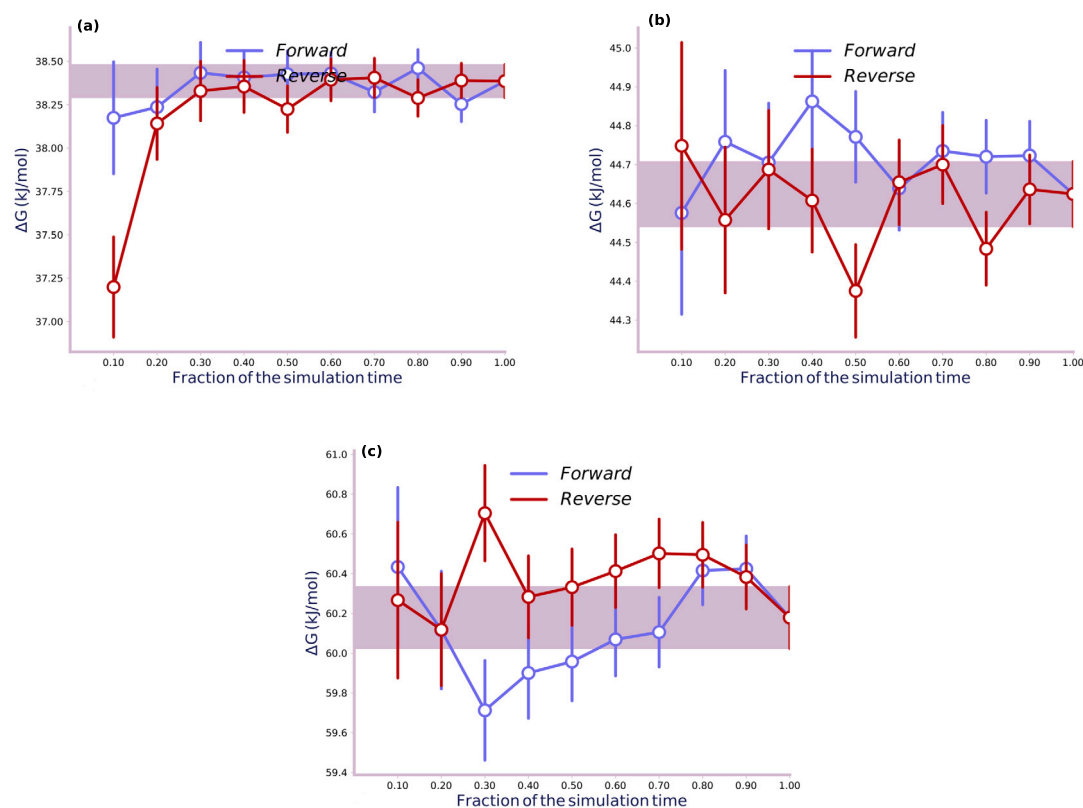


Fig. 3. Estimated value of free energy differences of (a) Gabapentin, (b) Paracetamol, and (c) Albuterol as a function of simulation time for both forward and reverse directions at 310 K temperature taking TIP3P water as solvent.

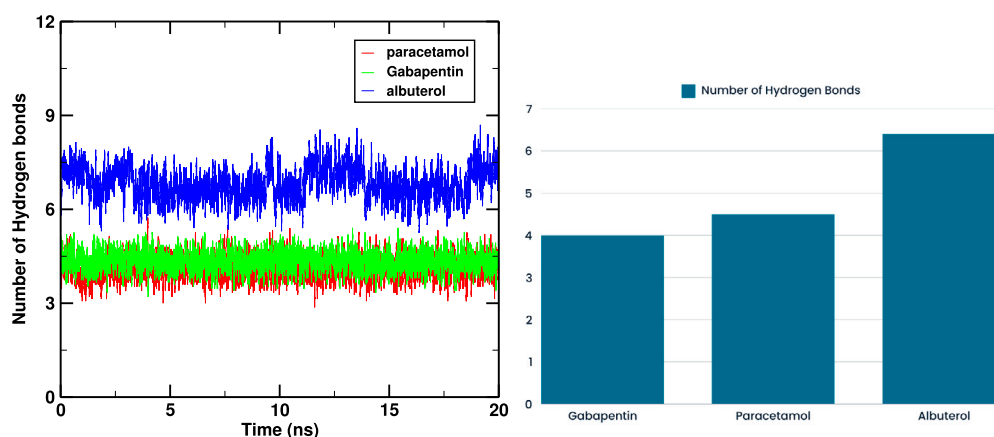


Fig. 4. Comparison of the time evolution of the number of hydrogen bonds between solvent and different solutes (left) and average number of hydrogen bonds (right) at 310 K temperature for coupling state represented by $\lambda = 0$.

The surface area of a molecule that is accessible to solvent molecules is known as SASA, and it is a fundamental idea in molecular simulations. It offers information on binding sites, hydrophobic/hydrophilic characteristics, protein folding, and molecular interactions. We have estimated SASA for each solute molecule at 310 K temperature for coupling state $\lambda = 0$ for TIP3P water which is represented in Fig. 5. Average SASA values for Gabapentin, Paracetamol, and Albuterol were found as 3.61 nm², 3.41 nm², and 4.93 nm² respectively. The SASA values for all drugs fluctuate around the mean value suggesting the simulation is stable over the simulation time. The SASA values increase with solute size. Since, $\Delta G = \gamma \times SASA + b$, where γ and b are constant [84–86]. This indicates larger SASA corresponds to large solvation-free energy. The higher value of SASA supports the higher solvation-free energy of albuterol than gabapentin and paracetamol.

4.3. Radial distribution function

The radial distribution function (RDF), denoted as $g(r)$, is a measure of the probability of finding a particle at a distance r from a reference particle relative to an ideal, uniform distribution. It is commonly used to investigate the structural properties of a system. It provides insight into a system's local structure and spatial arrangement of particles. RDF analysis also helps in understanding hydration shells around solutes, structural organization in complex fluids, and protein-ligand interactions.

The radial distribution function (RDF) plots illustrate how water molecules are distributed around different solute atoms in aqueous solutions of (a) Gabapentin, (b) Paracetamol, and (c) Albuterol. Here, the first peak for OW-OW in (a), (b), and (c) is 0.27 nm, and the correspond-

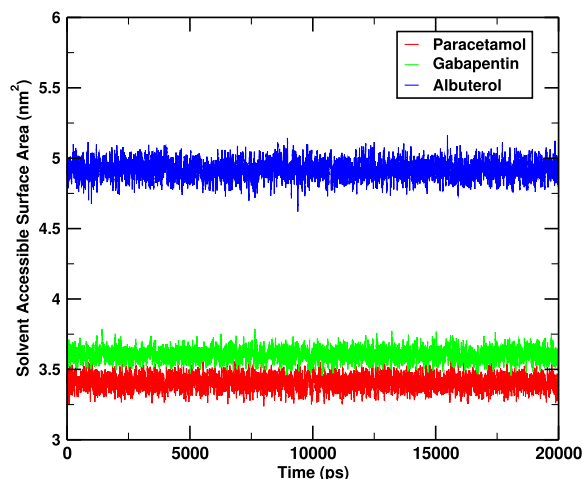


Fig. 5. Comparison of the time evolution of Solvent Accessible Surface Area (SASA) of different solute molecules at 310 K temperature for coupling state $\lambda = 0$ for TIP3P water.

ing value of $g(r)$ is 3.03. This sharp peak around 0.27 nm represents the first solvation shell, where the oxygen atoms in water molecules are most densely packed due to hydrogen bonding. It is observed that the excluded region (ER) values and the first peak position (FPP) are smaller than the van der Waals radius (0.355 nm [73]), suggesting that, in addition to the van der Waals potential, other interactions also influence the structural properties of the system [80,87]. For Gabapentin (a), the RDF of OW-OG (oxygen of water around Gabapentin's oxygen) has peaks at $r \approx 0.27$ nm with the corresponding value of $g(r) \approx 0.87$, indicating strong hydrogen bonding between water molecules and the oxygen atom of Gabapentin. Similarly, the OW-NG (nitrogen of Gabapentin) shows a first peak at $r \approx 0.29$ nm with the corresponding value of $g(r) \approx 1.15$, suggesting even stronger and more structured solvation around the nitrogen atom.

In contrast, in Paracetamol (b), OW-OP (oxygen of water around oxygen in Paracetamol) peaks at $r \approx 0.30$ nm with $g(r) \approx 1.43$, suggesting a more diffuse hydration shell around its hydroxyl and amide oxygen atoms. The OW-NP (oxygen of water around nitrogen in Paracetamol) rdf curve has a relatively flat peak at $r \approx 0.29$ nm with $g(r) \approx 0.42$, indicating minimal structured water clustering around the nitrogen site, likely due to steric hindrance and limited polarity compared to Gabapentin.

For Albuterol (c), OW-OA (oxygen of water around Albuterol's oxygen) has a first peak at $r \approx 0.28$ nm with the highest $g(r)$ value (≈ 1.56) among all the solutes (gabapentin and paracetamol). In contrast, OW-NA (oxygen of water around nitrogen in Albuterol) exhibits a smaller peak at $r \approx 0.27$ nm, indicating weaker solvation around the nitrogen

site, though slightly better than in the other two solutes. The RDF results suggest that the oxygen atoms of all three solutes form strong hydrogen bonds with water, with Albuterol exhibiting the most structured solvation shell, followed by Paracetamol and Gabapentin (Fig. 6).

4.4. Transport properties

In addition to the calculation of solvation-free energy, we also estimated the transport properties of drug molecules and the water model. To estimate the self-diffusion coefficients of the solute and solvent in the system using Einstein's relation (Eq. (5)), the mean square displacement (MSD) was plotted as a function of time on a logarithmic scale. Although the production run in the NVT ensemble was carried out for 1000 ns, the MSD vs. time graph was truncated to 5 ns to improve statistical accuracy. Fig. 7(a–d) shows log-log scale MSD versus time plots of SPC/E water, Gabapentin, paracetamol, and Albuterol respectively. At short timescales, the graph shows particle motion in molecular systems is a ballistic regime with a mean square displacement proportional to the square of time ($MSD \propto t^2$). Since collisions are minimal, this regime corresponds to particles moving freely with their initial velocities [88]. As molecular size increases, the ballistic regime becomes more evident [65]. Smaller molecules like SPC/E water quickly transition to the diffusive regime due to frequent collisions, whereas large molecules like gabapentin, paracetamol, and albuterol display a more prominent ballistic phase because of the slower dissipation of momentum due to less frequent collisions. As the system transits to a diffusive regime, MSD grows linearly with time ($MSD \propto t$), corresponding to the completely random collision-driven process of motion [66]. Figures (a–d) show that the transition for the diffusive regime for Gabapentin, Paracetamol, and Albuterol occurs around 120 ps, 50 ps, and 200 ps of simulation time, respectively.

The estimated values of the self-diffusion coefficients of drug molecules and the water model were obtained by linearly fitting the MSD vs. time graph. The estimated values of the Self-diffusion coefficient of Gabapentin, Paracetamol, Albuterol, and SPC/E water and their binary diffusion coefficients are presented in Table 2. The binary diffusion coefficients are calculated using Darken's relation in (6). The estimated values of the self-diffusion coefficient of water closely agree with the previously reported MD simulation data. The experimental data was obtained at 313 K, which suggests our estimation is accurate considering the effect of diffusion on temperature. Similarly, estimated values of self Diffusion for Gabapentin and Albuterol closely agree with experimental data with $\approx 10\%$ error value. However for Paracetamol, the experimental value of self-diffusion was obtained $0.664 \times 10^{-9} \text{ m}^2/\text{s}$ at 298 K. If we consider the temperature dependence effect on the diffusion coefficient, then we can assume that the obtained value of self-diffusion coefficient is reliable. Table 2 shows that the binary diffusion coefficients closely approximate the corresponding self-diffusion coefficient

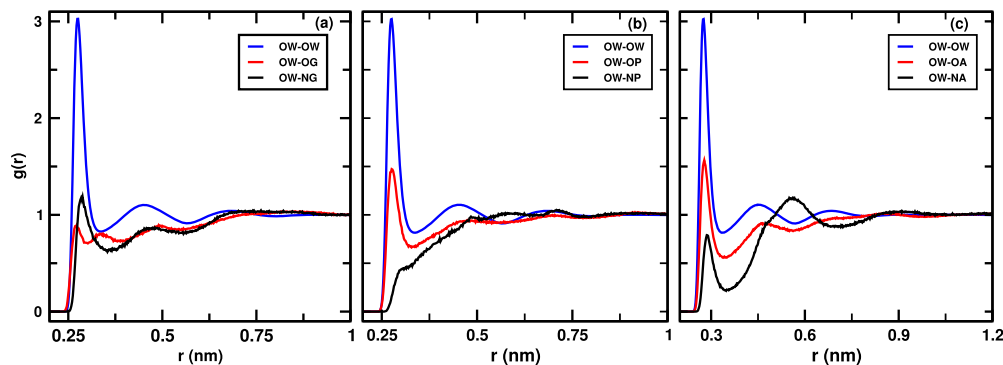


Fig. 6. Radial distribution function of aqueous solutions of (a) Gabapentin, (b) Paracetamol, and (c) Albuterol. Each subplot shows RDFs for oxygen of water (OW) around oxygen of water (OW-OW), oxygen of water around oxygen of the solute (OW-OG, OW-OP, OW-OA), and oxygen of water around nitrogen of the solute (OW-NG, OW-NP, OW-NA).

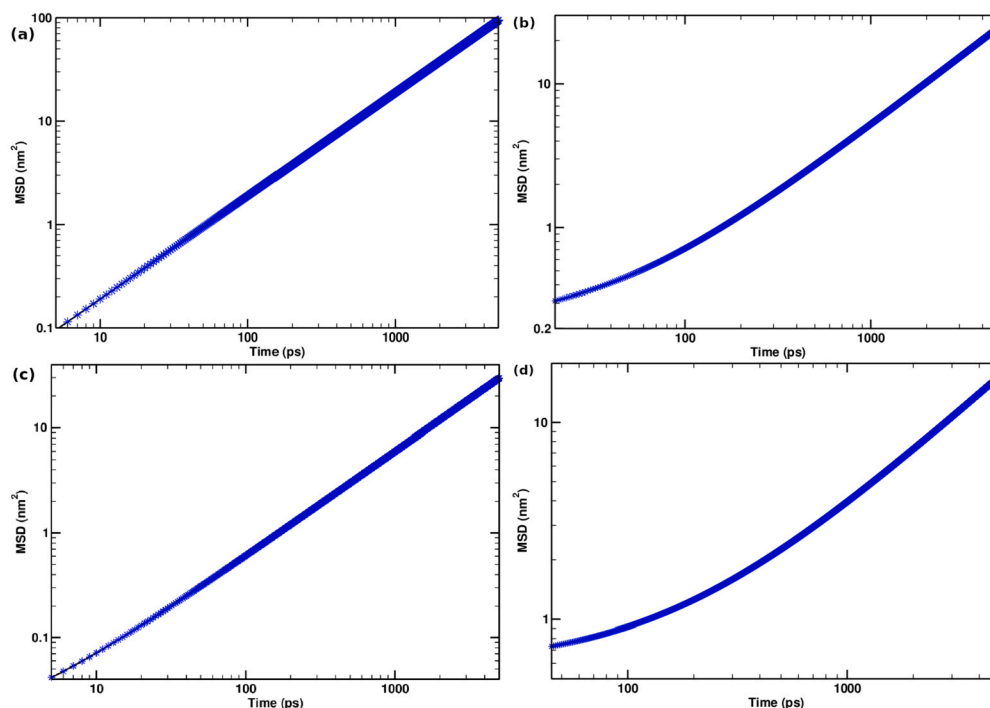


Fig. 7. MSD versus time plot in logarithmic scale for (a) SPC/E Water (b) Gabapentin (c) Paracetamol (d) Albuterol at temperature 310 K.

Table 2

The simulated values of self-diffusion coefficients as well as Binary diffusion coefficient at 310 K.

System	Self-Diffusion ($10^{-9} \text{ m}^2/\text{s}$)				Binary-Diffusion ($10^{-9} \text{ m}^2/\text{s}$)	
	MSD	Ref.	Expt.	% Error	MSD	
Gabapentin	0.83 ± 0.02	-	0.93 [89]	10.17	0.84	
Paracetamol	0.990 ± 0.02	-	0.664 [90]	-	0.992	
Albuterol	0.56 ± 0.02	-	0.62 [43]	9.68	0.56	
SPC/E water	3.11 ± 0.01	3.14 [37]	3.22 [91]	-		

Table 3

Estimated values of Self-Diffusion and Shear-Viscosity of SPC/E water model and solution at 310 K.

System	Self-Diffusion ($10^{-9} \text{ m}^2/\text{s}$)			Shear-Viscosity ($10^{-4} \text{ kg m}^{-1} \text{ s}^{-1}$)		
	Calculated	Ref.	Expt.	Calculated	Ref.	Expt.
SPC/E water	3.11 ± 0.01	3.14 [37]	3.22 [91]	6.19 ± 0.01	6.42 [42]	7.98 [92]
Gabapentin-Water	3.06 ± 0.01	-	-	6.82 ± 0.01	-	-
Paracetamol-Water	3.06 ± 0.01	-	-	5.17 ± 0.01	-	-
Albuterol-Water	3.08 ± 0.01	-	-	5.74 ± 0.02	-	-

values of the drug molecules, likely due to the low concentration of drug molecules in the system.

Along with the self and binary diffusion coefficients, we also calculated the shear viscosity using Einstein's relation (Eq. (7)). Table 3 presents the estimated self-diffusion coefficients and shear viscosity values for the SPC/E water model and drug-water solutions. The results indicate that the self-diffusion coefficients for the drug-water solutions are lower than those of pure water, reflecting a slower diffusion rate due to the presence of dissolved drug molecules. For the SPC/E water model, the estimated shear viscosity values closely align with both reference data and experimental results, accounting for temperature effects, as the reference values were measured at 300 K.

5. Conclusion & outlooks

In this work, we have presented the thermodynamic and transport properties of gabapentin, paracetamol, and albuterol by using classical molecular dynamics simulation. OPLS-AA force field and TIP3P,

redTIP4P/2005, and SPC/E water models were used during the simulation. The solvation-free energy of these drugs at 310 K temperature has been estimated using thermodynamic integration (TI) and free energy perturbation (FEP) techniques, including TI, TI-Cubic, BAR, and MBAR. Further, our study suggests that the contribution of Coulomb interaction to the free energy is greater than that of van der Waals interaction. To better understand the impact of the solvent environment on solvation-free energy at the molecular level, the SASA of the solute and the number of hydrogen bonds between the solute and solvent molecules have been calculated for the initial thermodynamic state using TIP3P as solvents. The average value of the SASA was 3.61 nm^2 , 3.41 nm^2 , and 4.93 nm^2 for gabapentin, paracetamol, and albuterol respectively. Hydrogen bond analysis and the value of SASA obtained also support the solvation of these drugs in water. Einstein's and Darken's relations have been used to estimate the self-diffusion coefficients of the solvent and solute, as well as their binary diffusion coefficient, at 310 K. The calculated values of the self-diffusion coefficient for gabapentin, paracetamol, and albuterol and the experimentally reported values in the literature

are in good agreement indicating the accuracy of the calculations. Furthermore, SPC/E water exhibits good consistency with reference data for self-diffusion. In addition, the shear viscosities of the solution and SPC/E water were calculated to provide insight into the properties of the fluid. The system's structural properties were analyzed using the radial distribution function (RDF). A crucial aspect of oral drug absorption is its solubility. Calculating the drug's solvation-free energy and absolute free energy in the crystal can be used to determine its solubility. Pharmaceuticals may greatly benefit from a comprehensive understanding of a drug's thermodynamic and transport properties.

CRedit authorship contribution statement

Ashok Chaudhary: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Bishnu Pandey:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Muskan Jalan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Shyam Prakash Khanal:** Writing – review & editing, Visualization, Validation, Investigation, Formal analysis, Data curation. **Narayan Prasad Adhikari:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.molliq.2025.127654>.

Data availability

Data will be made available on request.

References

- [1] J. Mao, L.L. Chen, Gabapentin in pain management, *Anesth. Analg.* 91 (3) (2000) 680–687.
- [2] M. Rose, P. Kam, Gabapentin: pharmacology and its use in pain management, *Anaesthesia* 57 (5) (2002) 451–462.
- [3] M.J. McLean, Gabapentin, *Epilepsia* 36 (1995) S73–S86.
- [4] C.P. Taylor, Mechanisms of action of gabapentin, *Rev. Neurol.* 153 (1997) S39–S45.
- [5] W.C. Risher, C. Eroglu, Emerging roles for $\alpha 2\delta$ subunits in calcium channel function and synaptic connectivity, *Curr. Opin. Neurobiol.* 63 (2020) 162–169.
- [6] G.J. Sills, The mechanisms of action of gabapentin and pregabalin, *Curr. Opin. Pharmacol.* 6 (1) (2006) 108–113.
- [7] A. Kukkar, A. Bali, N. Singh, A.S. Jaggi, Implications and mechanism of action of gabapentin in neuropathic pain, *Arch. Pharm. Res.* 36 (2013) 237–251.
- [8] Y. Maneuf, Z. Luo, K. Lee, $\alpha 2\delta$ and the mechanism of action of gabapentin in the treatment of pain, *Semin. Cell Dev. Biol.* 17 (2006) 565–570.
- [9] H.N. Bockbrader, D. Wesche, R. Miller, S. Chapel, N. Janiczek, P. Burger, A comparison of the pharmacokinetics and pharmacodynamics of pregabalin and gabapentin, *Clin. Pharmacokinet.* 49 (2010) 661–669.
- [10] H.N. Morse, Ueber eine neue darstellungsmethode der acetylamidophenole, *Ber. Dtsch. Chem. Ges.* 11 (1) (1878) 232–233.
- [11] Z. Iranmanesh-Zarandy, M. Dehestani, Molecular dynamics simulation of paracetamol drug adsorption on boron nitride nanotube: effects of temperature, nanotube length, diameter, and chirality, *ChemistrySelect* 4 (27) (2019) 7866–7873.
- [12] L. Ahmed, R. Omer, Computational study on paracetamol drug, *J. Phys. Chem. Funct. Mater.* 3 (1) (2020) 9–13.
- [13] U. Freo, C. Ruocco, A. Valerio, I. Scagnoli, E. Nisoli, Paracetamol: a review of guideline recommendations, *J. Clin. Med.* 10 (15) (2021) 3420.
- [14] G.G. Graham, K.F. Scott, Mechanism of action of paracetamol, *Am. J. Ther.* 12 (1) (2005) 46–55.
- [15] E. Ricciotti, G.A. FitzGerald, Prostaglandins and inflammation, *Arterioscler. Thromb. Vasc. Biol.* 31 (5) (2011) 986–1000.
- [16] N. Chandrasekharan, H. Dai, K.L.T. Roos, N.K. Evanson, J. Tomsik, T.S. Elton, D.L. Simmons, Cox-3, a cyclooxygenase-1 variant inhibited by acetaminophen and other analgesic/antipyretic drugs: cloning, structure, and expression, *Proc. Natl. Acad. Sci.* 99 (21) (2002) 13926–13931.
- [17] G.-W. An, H. Zhang, X.-L. Cheng, Q.-L. Zhuo, Y.-C. Lv, Electronic structure and hydrogen bond in the crystal of paracetamol drugs, *Struct. Chem.* 19 (2008) 613–617.
- [18] C. Wilson, Neutron diffraction of p-hydroxyacetanilide (paracetamol): libration or disorder of the methyl group at 100 K, *J. Mol. Struct.* 405 (2–3) (1997) 207–217.
- [19] S.-H. Lin, W. Cui, G.-L. Wang, S. Meng, Y.-C. Liu, H.-W. Jin, L.-R. Zhang, Y. Xie, Combined computational and experimental studies of molecular interactions of albuterol sulfate with bovine serum albumin for pulmonary drug nanoparticles, *Drug Des. Dev. Ther.* (2016) 2973–2987.
- [20] K. Makhoul, H.L. Weiner, S.J. Khoury, Potential of β 2-adrenoceptor agonists as add-on therapy for multiple sclerosis: focus on salbutamol (albuterol), *CNS Drugs* 16 (2002) 1–8.
- [21] I. Citaristi, World health organization—WHO, in: *The Europa Directory of International Organizations 2022*, Routledge, 2022, pp. 380–395.
- [22] N. Ullmann, S. Caggiano, R. Cutrera, Salbutamol and around, *Ital. J. Pediatr.* 41 (Suppl 2) (2015) A74.
- [23] D.R. Taylor, G.I. Town, G.P. Herbison, D. Boothman-Burrell, E.M. Flannery, B. Hancock, E. Harré, K. Laubscher, V. Linscott, C.M. Ramsay, et al., Asthma control during long term treatment with regular inhaled salbutamol and salmeterol, *Thorax* 53 (9) (1998) 744–752.
- [24] H.Y. Aboul-Enein, A.A. Al-Badr, S. Ibrahim, Salbutamol, *Analytical Profiles of Drug Substances*, vol. 10, Elsevier, 1981, pp. 665–689.
- [25] R.P. Bakale, S.A. Wald, H.T. Butler, Y. Gao, Y. Hong, X. Nie, C.M. Zepp, Albuterol: a pharmaceutical chemistry review of r-, s-, and rs-albuterol, *Clin. Rev. Allergy Immunol.* 14 (1996) 7–35.
- [26] A.S. Kesselheim, B. Wang, J.M. Franklin, J.J. Darrow, Trends in utilization of fda expedited drug development and approval programs, 1987–2014: cohort study, *BMJ* 351 (2015).
- [27] A.H. Price, S.P. Clissold, Salbutamol in the 1980s: a reappraisal of its clinical efficacy, *Drugs* 38 (1) (1989) 77–122.
- [28] A. Emeryk, J. Emeryk-Maksymiuk, Short-acting inhaled $\beta 2$ -agonists: why, whom, what, how?, *Adv. Respir. Med.* 88 (5) (2020) 443–449.
- [29] L. Marques, N. Vale, Unraveling the impact of salbutamol polytherapy: clinically relevant drug interactions, *Future Pharmacol.* 3 (1) (2023) 296–316.
- [30] S.E. Libretto, A review of the toxicology of salbutamol (albuterol), *Arch. Toxicol.* 68 (1994) 213–216.
- [31] C.H. Mehta, R. Narayan, U.Y. Nayak, Computational modeling for formulation design, *Drug Discov. Today* 24 (3) (2019) 781–788.
- [32] S. Hossain, A. Kabelev, A. Parrow, C.A. Bergström, P. Larsson, Molecular simulation as a computational pharmaceuticals tool to predict drug solubility, solubilization processes and partitioning, *Eur. J. Pharm. Biopharm.* 137 (2019) 46–55.
- [33] Y. Lee, S. Basith, S. Choi, Recent advances in structure-based drug design targeting class ag protein-coupled receptors utilizing crystal structures and computational simulations, *J. Med. Chem.* 61 (1) (2018) 1–46.
- [34] N. Hansen, W.F. Van Gunsteren, Practical aspects of free-energy calculations: a review, *J. Chem. Theory Comput.* 10 (7) (2014) 2632–2647.
- [35] R. Skyner, J. McDonagh, C. Groom, T.v. van Mourik, J. Mitchell, A review of methods for the calculation of solution free energies and the modelling of systems in solution, *Phys. Chem. Chem. Phys.* 17 (9) (2015) 6174–6191.
- [36] D.L. Mobley, P.V. Klimovich, Perspective: alchemical free energy calculations for drug discovery, *J. Chem. Phys.* 137 (23) (2012).
- [37] S.P. Khanal, N.P. Adhikari, Thermodynamic and transport properties of amoxicillin, *J. Mol. Liq.* 354 (2022) 118865.
- [38] T. Bhatta, P. Khanal, S.P. Khanal, N.P. Adhikari, Thermodynamics and transport properties of valine and cysteine peptides in water, *J. Mol. Liq.* 376 (2023) 121472.
- [39] Y. Nademi, S. Amjad Iranagh, A. Yousefpour, S.Z. Mousavi, H. Modarress, Molecular dynamics simulations and free energy profile of paracetamol in dppe and dmpe lipid bilayers, *J. Chem. Sci.* 126 (2014) 637–647.
- [40] M.A. Bellucci, G. Gobbo, T.K. Wijethunga, G. Ciccotti, B.L. Trout, Solubility of paracetamol in ethanol by molecular dynamics using the extended Einstein crystal method and experiments, *J. Chem. Phys.* 150 (9) (2019) 094107.
- [41] Y. Wang, S. Du, S. Wu, L. Li, D. Zhang, B. Yu, L. Zhou, H. Kiflegiorgis Bekele, J. Gong, Thermodynamic and molecular investigation into the solubility, stability and self-assembly of gabapentin anhydrate and hydrate, *J. Chem. Thermodyn.* 113 (2017) 132–143.
- [42] B. Hess, Determining the shear viscosity of model liquids from molecular dynamics simulations, *J. Chem. Phys.* 116 (1) (2002) 209–217.
- [43] S. Miyamoto, K. Shimono, Molecular modeling to estimate the diffusion coefficients of drugs and other small molecules, *Molecules* 25 (22) (2020) 5340.
- [44] X. Wang, Y. Liu, J. Xu, S. Li, F. Zhang, Q. Ye, X. Zhai, X. Zhao, Molecular dynamics study of stability and diffusion of graphene-based drug delivery systems, *J. Nanomater.* 2015 (1) (2015) 872079.
- [45] A.A. Noyes, W.R. Whitney, The rate of solution of solid substances in their own solutions, *J. Am. Chem. Soc.* 19 (12) (1897) 930–934.
- [46] Y. Gao, K.W. Olsen, Molecular dynamics of drug crystal dissolution: simulation of acetaminophen form I in water, *Mol. Pharm.* 10 (3) (2013) 905–917.

- [47] D. Shivakumar, Y. Deng, B. Roux, Computations of absolute solvation free energies of small molecules using explicit and implicit solvent model, *J. Chem. Theory Comput.* 5 (4) (2009) 919–930.
- [48] M. Karplus, J.A. McCammon, Molecular dynamics simulations of biomolecules, *Nat. Struct. Biol.* 9 (9) (2002) 646–652.
- [49] E.A. Rifai, V. Ferrario, J. Pleiss, D.P. Geerke, Combined linear interaction energy and alchemical solvation free-energy approach for protein-binding affinity computation, *J. Chem. Theory Comput.* 16 (2) (2020) 1300–1310.
- [50] R. Freitas, M. Asta, M. De Koning, Nonequilibrium free-energy calculation of solids using lammmps, *Comput. Mater. Sci.* 112 (2016) 333–341.
- [51] D.L. Beveridge, F.M. Dicapua, Free energy via molecular simulation: applications to chemical and biomolecular systems, *Annu. Rev. Biophys. Biophys. Chem.* 18 (1) (1989) 431–492.
- [52] T. Hou, J. Wang, Y. Li, W. Wang, Assessing the performance of the mm/pbsa and mm/gbsa methods. 1. The accuracy of binding free energy calculations based on molecular dynamics simulations, *J. Chem. Inf. Model.* 51 (1) (2011) 69–82.
- [53] M. Mezei, D. Beveridge, Free energy simulations, *Ann. N.Y. Acad. Sci.* 482 (1) (1986) 1–23.
- [54] J. Ho, M.Z. Ertem, Calculating free energy changes in continuum solvation models, *J. Phys. Chem. B* 120 (7) (2016) 1319–1329.
- [55] P. Virmann, M. Müller, Calculation of free energy through successive umbrella sampling, *J. Chem. Phys.* 120 (23) (2004) 10925–10930.
- [56] C. Chipot, A. Pohorille, *Free Energy Calculations*, vol. 86, Springer, 2007.
- [57] Y. Levy, J.N. Onuchic, Water mediation in protein folding and molecular recognition, *Annu. Rev. Biophys. Biomol. Struct.* 35 (1) (2006) 389–415.
- [58] P. Kollman, Free energy calculations: applications to chemical and biochemical phenomena, *Chem. Rev.* 93 (7) (1993) 2395–2417.
- [59] S. Wan, R.H. Stote, M. Karplus, Calculation of the aqueous solvation energy and entropy, as well as free energy, of simple polar solutes, *J. Chem. Phys.* 121 (19) (2004) 9539–9548.
- [60] R.W. Zwanzig, High-temperature equation of state by a perturbation method. I. Non-polar gases, *J. Chem. Phys.* 22 (8) (1954) 1420–1426.
- [61] M.E. Tuckerman, *Statistical Mechanics: Theory and Molecular Simulation*, Oxford University Press, 2023.
- [62] J.G. Kirkwood, Statistical mechanics of fluid mixtures, *J. Chem. Phys.* 3 (5) (1935) 300–313.
- [63] J. Crank, *The Mathematics of Diffusion*, Clarendon, Oxford, England, 1975.
- [64] S. Pokharel, N. Pantha, N. Adhikari, Diffusion coefficients of nitric oxide in water: a molecular dynamics study, *Int. J. Mod. Phys. B* 30 (27) (2016) 1650205.
- [65] D. Frenkel, B. Smit, M.A. Ratner, *Understanding Molecular Simulation: From Algorithms to Applications*, vol. 2, Academic Press, San Diego, 1996.
- [66] M.P. Allen, D.J. Tildesley, *Computer Simulation of Liquids*, Oxford University Press, 2017.
- [67] V. Ballenegger, S. Picaud, C. Toubin, Molecular dynamics study of diffusion of formaldehyde in ice, *Chem. Phys. Lett.* 432 (1–3) (2006) 78–83.
- [68] L.S. Darken, Diffusion, mobility and their interrelation through free energy in binary metallic systems, *Trans. AIME* 175 (1948) 184–201.
- [69] K. Rah, B.C. Eu, Relation of shear viscosity and self-diffusion coefficient for simple liquids, *Phys. Rev. E* 60 (4) (1999) 4105.
- [70] E.J. Maginn, R.A. Messerly, D.J. Carlson, D.R. Roe, J.R. Elliot, Best practices for computing transport properties 1. Self-diffusivity and viscosity from equilibrium molecular dynamics [article v1.0], *Living J. Comput. Mol. Sci.* 1 (1) (2019) 6324.
- [71] D. Van Der Spoel, E. Lindahl, B. Hess, G. Groenhof, A.E. Mark, H.J. Berendsen, Gromacs: fast, flexible, and free, *J. Comput. Chem.* 26 (16) (2005) 1701–1718.
- [72] W.L. Jorgensen, D.S. Maxwell, J. Tirado-Rives, Development and testing of the opls all-atom force field on conformational energetics and properties of organic liquids, *J. Am. Chem. Soc.* 118 (45) (1996) 11225–11236.
- [73] M. Abraham, D. van der Spoel, E. Lindahl, development team, GROMACS user manual version 5.1.1, www.gromacs.org, 2015.
- [74] P. Mark, L. Nilsson, Structure and dynamics of the tip3p, spc, and spc/e water models at 298 K, *J. Phys. Chem. A* 105 (43) (2001) 9954–9960.
- [75] J.L. Abascal, C. Vega, A general purpose model for the condensed phases of water: Tip4p/2005, *J. Chem. Phys.* 123 (23) (2005).
- [76] B. Hess, H. Bekker, H.J. Berendsen, J.G. Fraaije, Lincs: a linear constraint solver for molecular simulations, *J. Comput. Chem.* 18 (12) (1997) 1463–1472.
- [77] P.V. Klimovich, M.R. Shirts, D.L. Mobley, Guidelines for the analysis of free energy calculations, *J. Comput.-Aided Mol. Des.* 29 (2015) 397–411.
- [78] H.J. Berendsen, J.-R. Grigera, T.P. Straatsma, The missing term in effective pair potentials, *J. Phys. Chem.* 91 (24) (1987) 6269–6271.
- [79] R.P. Koirala, H.P. Bhusal, S.P. Khanal, N.P. Adhikari, Effect of temperature on transport properties of cysteine in water, *AIP Adv.* 10 (2) (2020).
- [80] S.P. Khanal, Y.P. Kandel, N.P. Adhikari, Transport properties of zwitterion glycine, diglycine, and triglycine in water, *AIP Adv.* 9 (6) (2019).
- [81] C.H. Bennett, Efficient estimation of free energy differences from Monte Carlo data, *J. Comput. Phys.* 22 (2) (1976) 245–268.
- [82] A.V. Marenich, C.P. Kelly, J.D. Thompson, G.D. Hawkins, C.C. Chambers, D.J. Giesen, P. Winget, C.J. Cramer, D.G. Truhlar, Minnesota solvation database (mnsol) version 2012, Retrieved from the Data Repository for the University of Minnesota 10 (2020).
- [83] A.S. Mey, B.K. Allen, H.E.B. Macdonald, J.D. Chodera, D.F. Hahn, M. Kuhn, J. Michel, D.L. Mobley, L.N. Naden, S. Prasad, et al., Best practices for alchemical free energy calculations [article v1.0], *Living J. Comput. Mol. Sci.* 2 (1) (2020).
- [84] D. Sitkoff, K.A. Sharp, B. Honig, Accurate calculation of hydration free energies using macroscopic solvent models, *J. Phys. Chem.* 98 (7) (1994) 1978–1988.
- [85] J. Zhang, H. Zhang, T. Wu, Q. Wang, D. Van Der Spoel, Comparison of implicit and explicit solvent models for the calculation of solvation free energy in organic solvents, *J. Chem. Theory Comput.* 13 (3) (2017) 1034–1043.
- [86] C. Tan, Y.-H. Tan, R. Luo, Implicit nonpolar solvent models, *J. Phys. Chem. B* 111 (42) (2007) 12263–12274.
- [87] S. Pokharel, N. Aryal, B.R. Niraula, A. Subedi, N.P. Adhikari, Transport properties of methane, ethane, propane, and n-butane in water, *J. Phys. Commun.* 2 (6) (2018) 065003.
- [88] A. Rahman, Correlations in the motion of atoms in liquid argon, *Phys. Rev.* 136 (2A) (1964) A405.
- [89] J. Madan, G. Chawla, V. Arora, R. Malik, A.K. Bansal, Unbiased membrane permeability parameters for gabapentin using boundary layer approach, *AAPS J.* 7 (2005) E224–E230.
- [90] A.C. Ribeiro, M.C. Barros, L.M. Verissimo, C.I. Santos, A.M. Cabral, G.D. Gaspar, M.A. Estesio, Diffusion coefficients of paracetamol in aqueous solutions, *J. Chem. Thermodyn.* 54 (2012) 97–99.
- [91] M. Holz, S.R. Heil, A. Sacco, Temperature-dependent self-diffusion coefficients of water and six selected molecular liquids for calibration in accurate 1h nmr pfg measurements, *Phys. Chem. Chem. Phys.* 2 (20) (2000) 4740–4742.
- [92] D.R. Lide, *CRC Handbook of Chemistry and Physics*, vol. 85, CRC Press, 2004.
- [93] Kalachand Mahali, Partha Sarathi Guin, Sanjay Roy, Bijoy Krishna Dolui, Solubility and solute–solvent interaction phenomenon of succinic acid in aqueous ethanol mixtures, *J. Mol. Liq.* 229 (2017) 172–177.